Fiscal Year:	FY 2007	**Task Last Updated:**	FY 05/11/2007
PI Name:	Jenkins, James T. Ph.D.		
Project Title:	Particle Segregation in Collisional Shearing Flows		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Granular flows		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	14853	**Congressional District:**	22
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	OTHER
Start Date:	05/28/2000	**End Date:**	12/31/2006
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:	NOTE: Received NCE to 12/31/2006, per N. Hall (9/06)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Louge, Michel (Cornell University)		
Grant/Contract No.:	NCC3-797		
Performance Goal No.:	1.23		
Performance Goal Text:	Pursue ground-breaking basic research in fluid physics and transport phenomena to provide fundamental understanding of natural phenomena affected by gravity for increased efficiency and effectiveness of space-based and industrial processes.		

	This research concerned flowing granular materials and involved the development of ways to predict the behavior of flows and the means to control them. Granular flows are common in industrial processes, mining operations, and in nature. In general, they are poorly understood. The research treats flows in which the particles interact through colli-sions rather than enduring contacts. Such flows are expected to be important in materials processing activities carried out in space and in mining operations on the surface of the Moon and Mars. The specific phenomenon of interest in the research is the segregation of the particles in a collisional flow due to differences in their size and/or mass. In many industrial processes a homogeneous aggregate is desired; in these, segregation is undesirable. However, in the mining industry, segregation is exploited in sorting and crushing operations. Because seg-regation is not well understood, attempts to suppress it or exploit it proceed on an ad hoc basis and are expensive. In systems that do not involve much agitation of the grains, several mechanisms that in-volve gravity have been identified as leading to segregation. However, in highly agitated flows there is a mechanism independent of gravity that is available to drive segregation. This is associated with spatial gradients in the energy of the velocity fluctuations of the grains. Collisional interactions between and among different types of grains require that, in general, differences in their concentrations exist to balance differences in particle fluc-tuation energy. This segregation mechanism is often masked by gravitational segregation mechanisms on Earth. It is expected to be of equal importance to gravitational segregation in the reduced gravity on Mars and to be the dominant mechanism for segregation on the Moon. It is the only mechanism for segregation in space.
Task Description:	The segregation of colliding particles of different size and mass was to be studied on the International Space Station in an axisymmetric shear cell in which the flow is created by the relative motion of bumpy boundaries of a cylindrical annulus. In this cell, the profile of particle agitation across the cell is controlled by employing boundaries with different bumpiness. The particle segregation is observed using digital video, image analysis, and sophisticated particle tracking algorithms. Two basic systems were to be examined: in one, the spheres are of equal size but differ in mass; in the other, they are of equal mass but differ in size. The observations were to be compared to results of simulations and the predictions of theory to establish their respective limits and suggest possible improve-ments. Unfortunately, due to a change in priorities at the Space Agency, funding for the experiment was withdrawn. An axisymmetric prototype shear cell was designed and built to study granular flows in microgravity. The cell incorporated two cylindrical boundaries that rotated in opposite directions. Both boundaries were fitted with appropriately designed bumps to achieve predetermined gradients in the fluctuation energy of the grains. These gradients produced gradients in the concentration of the two types of particles in the cell. The prototype shear cell was flown on the KC135 microgravity aircraft. A single system was examined in which the spheres differed in both size and mass. The particle segregation was observed using digital video, image analysis, and sophisticated particle tracking algorithms. The observations were compared with the results of simulations and the predictions of theory. Binary impact parameters needed as input to the simulations and the theory were measured for collisions between two spheres and between a sphere and a boundary. The agreement between the profiles predicted by the theory and those measured in the ex-periment were, in general, good.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The study of a segregation mechanism that is especially important in reduced gravity should have benefited mining and materials-handling activities associated with in-situ resource utilization applications on Mars and the Moon. It should also have assisted in the interpretation of geologic deposits, particularly in low gravity. It would also have eventually benefited the design of manufacturing operations and in-space fabrication technologies in zero gravity in support of exploration.
Task Progress:	Flow development in an early non-axisymmetric shear cell was predicted and the importance of the bulk viscosity to this flow development highlighted. Boundary conditions at a bumpy, frictional boundary were derived using the expedient of superposing those for bumpy, frictionless boundaries and those for flat frictional boundaries. A simplified kinetic theory for the segregation of a binary mixture of nearly elastic spherical particles in a collisional shearing flow was developed. The predictions of this theory were compared with the results of numerical simulations and physical experiments carried out in a circular shear cell in microgravity on the KC-135. Friction was incorporated into the collisional interactions between spheres and circular disks and a correction to the coefficient of restitution in the kinetic theory for frictionless disks was calculated to include the translational energy lost to frictional sliding and the rotational velocity fluctuations. A simplified kinetic theory was developed for segregation of disks or spheres under gravity and applied to predict profiles of concentration and the strength of the velocity fluctuations in a binary mixture of spheres above a vibrating horizontal plate. The predictions were compared with the results of numerical simulations and experiments. The influence of the departure from equipartition of energy on segregation in binary mixtures under gravity was evaluated for systems of spheres and systems of circular disks in situations in which one phase was dilute and the other was dense. Finally, an attempt was made to emphasize the importance of research on granular materials to the space exploration program.
Bibliography Type:	Description: (Last Updated: 11/27/2020)
Articles in Peer-reviewed Journals	Yoon DK, Jenkins JT. "The influence of different species' granular temperatures on segregation in a binary mixture of dissipative grains." Physics of Fluids. 2006 Jul;18(7):073303. http://dx.doi.org/10.1063/1.2219437 , Jul-2006

Articles in Peer-reviewed Journals	Wildman RD, Jenkins JT, Krouskop PE, Talbot J. "A comparison of the predictions of a simple kinetic theory with experimental and numerical results for a vibrated granular bed consisting of nearly elastic particles of two sizes." Physics of Fluids. 2006 Jul;18(7):073301. http://dx.doi.org/10.1063/1.2210500 , Jul-2006
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Articles in Peer-reviewed Journals	Yoon DK, Jenkins JT. "Kinetic theory for identical, frictional, nearly elastic disks." Physics of Fluids. 2005 Aug;17(8):083301. http://dx.doi.org/10.1063/1.2000768 , Aug-2005
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Papers from Meeting Proceedings	Yoon DK, Jenkins JT. "Segregation in a sheared binary mixture of nearly elastic, smooth disks under gravity." Fifth International Conference on the Micromechanics of Granular Media, Powders and Grains 2005, Stuttgart, July 18-22, 2005. Powders and Grains 2005. Ed. by R.García-Rojo, H.J. Herrmann & S. McNamara. Rotterdam : Balkema, 2005. p. 1135-1139. , Jul-2005
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